scientific reports

Check for updates

OPEN

Enhanced YOLOv8 for efficient road damage detection with spatial-channel reconstruction and multi-scale attention

Zhipeng Tang[1], Hua Wang[1] & Rapeeporn Chamchong[2]✉

Accurate and efficient detection of road damage is essential for maintaining road safety and supporting intelligent transportation systems. While recent approaches leverage deep learning-based object detection frameworks, they often struggle with high computational demands and suboptimal feature extraction in complex environments. To address these challenges, we propose an enhanced object detection network for road damage detection based on the YOLOv8 architecture. Specifically, we integrate the Spatial and Channel Reconstruction Convolution (SCConv) module into the backbone to reduce feature redundancy while improving spatial and channel representation through a separation-reconstruction strategy. To enhance multi-scale feature fusion, we incorporate the Efficient Multi-Scale Attention (EMA) module into the neck, enabling adaptive spatial-channel attention without introducing significant computational overhead. Extensive benchmark comparison identifies YOLOv8l as a strong baseline for road damage detection. Building upon this, we embed SCConv within the original convolutional modules to achieve a lightweight network and enhance feature representation. We further explore multiple EMA integration strategies to identify an effective model configuration. Experimental results demonstrate that our best-performing model achieves higher detection accuracy than the baseline model, while maintaining computational efficiency.

Road infrastructure is the backbone of modern transport, supporting economic growth and social connectivity. Despite their importance, road surfaces inevitably deteriorate due to traffic loads, weather conditions, and aging[1]. This degradation manifests itself as potholes, cracks, and pavement wear that compromise road integrity. Early detection and timely repair of road damage yield substantial benefits for both communities and governments. However, road damage presents significant detection challenges due to the diverse nature of surface defects, which range from subtle cracks to major structural failures. Environmental conditions such as variable lighting, shadows, and weather can mask damage or create false indicators. The large scale of road networks compounds these difficulties. Human inspection presents numerous challenges due to its time-consuming nature, labor intensity, and inherent subjectivity, which often leads to inconsistent condition assessments[2,3]. These challenges drive the development of reliable and rapid assessment methods that can accurately identify various damage types across extensive networks under diverse conditions.

Deep learning-based object detection algorithms offer compelling advantages for road damage detection. Despite recent advances, numerous deep learning models exhibit substantial computational complexity and large parameters, limiting their suitability for deployment on mobile or resource-constrained embedded systems[4]. This highlights the need for lightweight architectures with faster inference to ensure practical usability in real-world road damage detection scenarios. In this paper, we propose an enhanced YOLOv8-based network architecture for road damage detection, incorporating Spatial and Channel Reconstruction Convolution (SCConv)[5] and Efficient Multi-Scale Attention (EMA)[6] mechanisms. Specifically, SCConv is designed to reduce feature redundancy and improve representation by sequentially applying spatial and channel reconstruction, which suppress spatial and channel redundancies via separate-and-reconstruct and split-transform-fuse strategies, respectively. We employ SCConv in place of traditional convolutions to reduce model size and computational complexity for faster inference, while enhancing feature representation and extraction. The EMA module integrates both channel and spatial attention mechanisms to adaptively weight features across different levels, further enhancing detection accuracy. We conduct experiments on different layers of the YOLOv8 network to identify the optimal configuration. In summary, the main contributions of this paper can be concluded as follows:

[1]College of Computer and Data Science, Putian University, Putian 351160, China. [2]Faculty of Informatics, Mahasarakham University, Maha Sarakham 44150, Thailand. ✉email: rapeeporn.c@msu.ac.th

- A comprehensive evaluation of representative benchmark object detectors, including YOLOv5, YOLOv8, YOLOv11, RT-DETRv2, RT-DETRv3, and EfficientDet, identifying YOLOv8l as a strong baseline for road damage detection;
- The novel integration of SCConv in the backbone and EMA in the neck of YOLOv8l, aiming at improving feature extraction and fusion capabilities in a lightweight architecture;
- The identification of an optimal configuration through extensive experimentation, achieving superior performance over the baseline model.

Related work

Road damage detection has emerged as a significant research area in computer vision and civil infrastructure monitoring. This task requires low-latency inference, and thus emphasizes lightweight network design to enable deployment on resource-constrained edge devices. Recent integration of attention mechanisms further enhances performance by allowing models to focus on relevant image regions while suppressing distracting elements.

Real-time object detectors

Real-time object detectors are designed to balance detection accuracy and inference speed. These detection frameworks are typically categorized into one-stage and two-stage detectors. One-stage detectors, represented by You Only Look Once (YOLO)[7], Single Shot MultiBox Detector (SSD)[8], RetinaNet[9], Real-Time Detection Transformer (RT-DETR)[10] and Fully Convolutional One-Stage Object Detection (FCOS)[11], execute localization and classification simultaneously, delivering efficient processing speeds conducive to real-time monitoring systems. Conversely, Two-stage detectors, exemplified by the R-CNN[12] family, first generate region proposals that potentially contain damage instances and then classify them, which often improves accuracy but increases inference time.

YOLO

YOLO has evolved remarkably since its first introduction in 2016, with each iteration bringing substantial improvements to the architecture. YOLOv8's architecture represents a significant advancement. As illustrated in Fig. 1, Its backbone utilizes an improved CSPDarknet with cross-stage partial connections, effectively balancing computational complexity and feature extraction capabilities. Unlike earlier versions, YOLOv8 completely abandons anchor boxes in favor of an anchor-free detection paradigm, which simplifies the prediction process. The neck employs an enhanced path aggregation network with feature pyramid network (PAN-FPN) built using the cross-stage partial (CSP) design for feature fusion, creating multi-scale feature representations that effectively integrate contextual information with spatial details. The detection head features a decoupled design that separates bounding box prediction and object classification.

Lightweight YOLO-based detectors

Recent YOLO-based lightweight designs for road damage detection mainly focus on (i) modular replacements for feature fusion and detection heads, (ii) structured compression, and (iii) hardware-oriented optimization. Li and Zhang[13] proposed BSE-YOLO, which enhances YOLOv8n with BiFPN-based feature fusion and lightweight

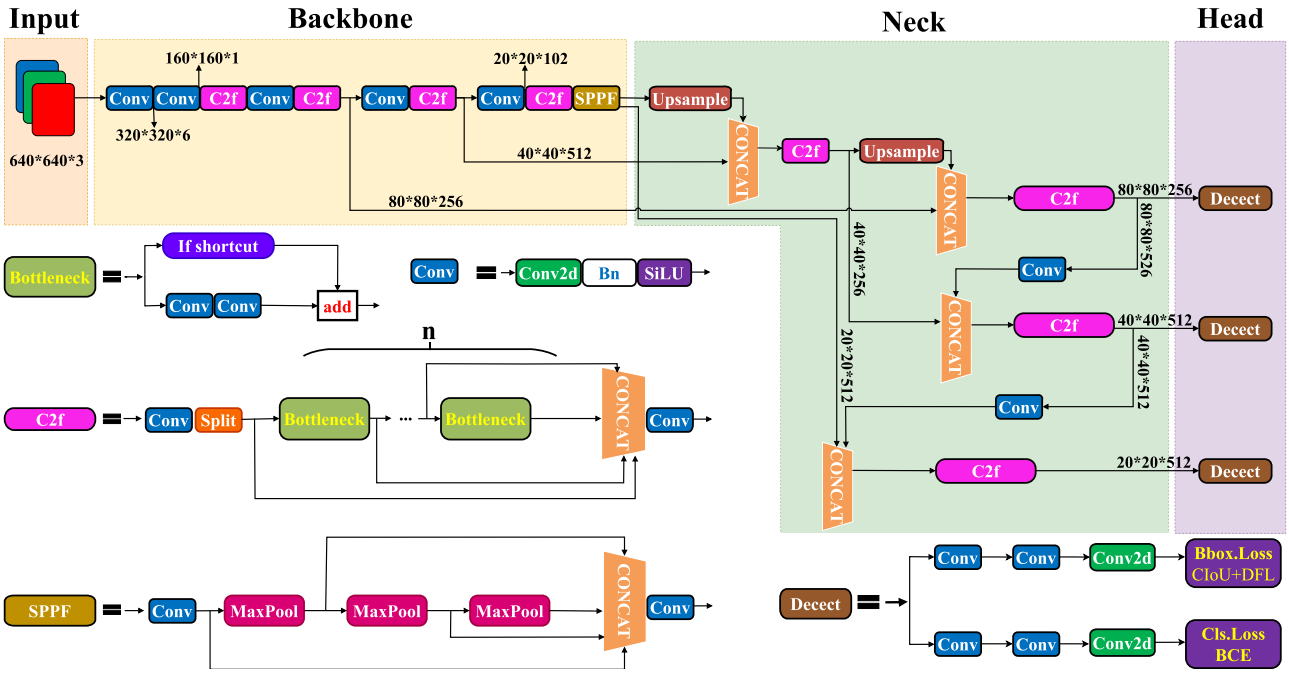


Fig. 1. YOLOv8 network architecture diagram.

redesign by replacing the neck block with SC2f, and adopting SEAHead to reduce computation while maintaining competitive detection performance. Lin and Pan[14] developed YOLO-ROC by combining a multi-scale enhancement module (BMS-SPPF) with hierarchical channel compression, substantially shrinking parameters and GFLOPs without sacrificing accuracy. Lan et al.[15] introduced a C2f-faster module in the neck and an LS-detect head of LRDD-YOLOv8 using group normalization and shared convolution to reduce computational cost. Beyond fusion and compression, Mao et al.[16] introduced YOLO-R, which incorporates dynamic receptive-field modeling in C2f via SAC embedding and adopts a GSConv-based slim-neck for lightweight designs, with ADown to mitigate information loss during downsampling. YOLOv8n-SBP[17] combines attention-enhanced backbone features (SENet), bidirectional fusion (BiFPN), and a parameter-sharing SCConv-empowered head to improve representation under constrained budgets. Yu et al.[18] introduced Mobile-YOLO, which adopts mobile-friendly blocks and lightweight attention (UIB/MQA) with NAS-based tuning to enable real-time crack detection on embedded devices. Reducing model parameters and computational complexity can improve inference speed; however, this often comes at the cost of decreased accuracy. Therefore, balancing the trade-off between lightweight design and detection performance is another key focus of this study.

Attention based detection methods

The introduction of attention mechanisms has significantly improved the performance of road damage detection models. These mechanisms allow the network to focus on the most relevant features in an image and enhance both the accuracy and robustness of the detection process. Liu et al.[19] developed an Efficient Dense Attention Fusion Network, which integrates a dense attention fusion module and channel correlation loss (CC-loss) to enhance feature separability, ultimately improving detection accuracy on the RDDA and RDD2022 datasets. Pan et al.[20] proposed a real-time dynamic scale-aware fusion network that introduces a Flexible Attention (FA) module combining DCNv2 and Triple Attention to enhance road-distress features under complex backgrounds. They further designed a Dynamic Scale-Aware Fusion (DSAF) strategy that applies FA across multi-scale sub-features and concatenates them for adaptive multi-scale representation. Wang et al.[21] integrated the EMA module into YOLOv8s to strengthen multi-scale feature fusion and improve generalization across diverse road conditions. Furthermore, they adopted a lightweight backbone and replaced CIoU with SIoU for bounding-box regression, achieving a better accuracy–efficiency trade-off. To address CNNs' limitations in modeling long-range dependencies, subsequent work focused on Swin-Transformer[22] integration: Irsal et al.[23] incorporated it into YOLOv7 to create YOLOv7-Swin, which outperforms standard YOLOv7 and YOLOv7x, while Roy and Bhaduri[24] proposed DenseSPH-YOLOv5 (fusing DenseNet with Swin-Transformer Prediction Heads) to enhance damage localization and balance detection speed and accuracy. More recently, Zhao et al.[25] proposed an improved RT-DETR for road crack detection by introducing a PCAM backbone (PKI C2f Attention Module) to enhance multi-scale perception, and by replacing Multi-Head Self-Attention with Cascaded Group Attention (CGA) to perform hierarchical feature aggregation with reduced computational cost.

Methods

SCConv structure and mechanism

SCConv is an efficient convolutional module designed to mitigate feature redundancy in deep convolutional neural networks. SCConv introduces two consecutive components: the Spatial Reconstruction Unit (SRU) and the Channel Reconstruction Unit (CRU). The SRU focuses on suppressing spatial redundancy by separating and reconstructing feature maps, while the CRU addresses channel redundancy using a split-transform-fuse strategy. These two units are sequentially arranged to enhance feature representation while significantly reducing computational overhead.

The SRU is designed to refine spatial feature representations by distinguishing informative regions from redundant ones and reconstructing them to improve feature expressiveness. The SRU consists of two key operations: separation and reconstruction. As depicted in Fig. 2, SRU employs group normalization (GN) to assess the informativeness of different spatial locations. Given an input feature map $X \in \mathbb{R}^{N\times C\times H\times W}$, where N is the batch size, C is the number of channels, and $H \times W$ represents the spatial dimensions, the normalization process is expressed as:

$$X_{out} = GN(X) = \gamma \frac{X-\mu}{\sqrt{\sigma^2+\varepsilon}} + \beta \tag{1}$$

where μ and σ are the mean and standard deviation, respectively, while γ and β are learnable affine transformation parameters. ϵ is a small positive constant added for the sake of division stability. The scaling factor γ serves as an indicator of spatial information richness, where higher values correspond to more spatially informative regions.

To effectively separate these feature maps, SRU first normalizes the correlation weights across all channels, ensuring that the relative importance of each feature map is accounted for. These normalized correlation weights (W_γ) are then processed through a sigmoid activation function, which maps them to the (0,1) range. A gating mechanism sets weights above 0.5 to 1 (informative feature maps, W_1) and those below 0.5 to 0 (less informative maps, W_2), effectively filtering redundant features. Finally, the input features X are multiplied by W_1 and W_2, producing two weighted feature sets: X_1^ω (informative features with expressive spatial content) and X_2^ω (less informative features with little or no useful information).

After separation, SRU reconstructs the feature maps to mitigate information loss while reinforcing the most relevant spatial features. Instead of simply discarding redundant features, a cross-reconstruction strategy is employed, where informative and less informative features are recombined interactively:

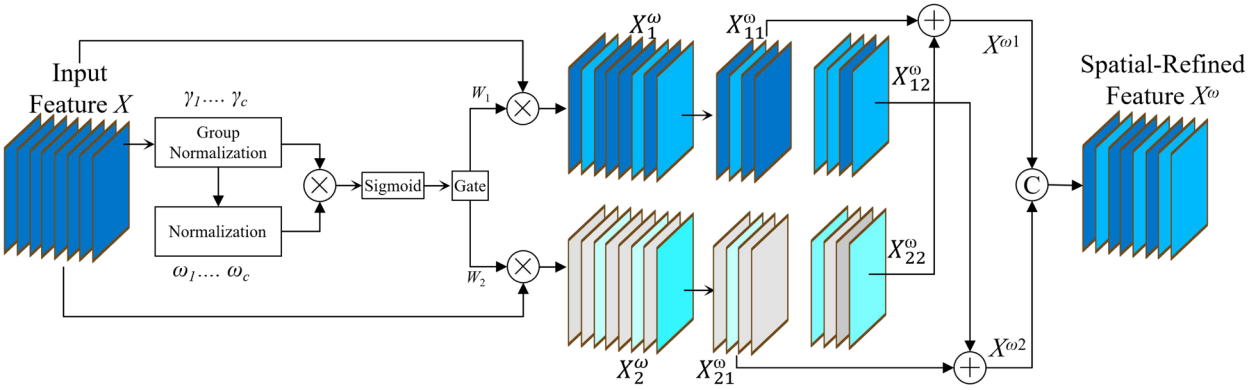


Fig. 2. The architecture of spatial reconstruction unit.

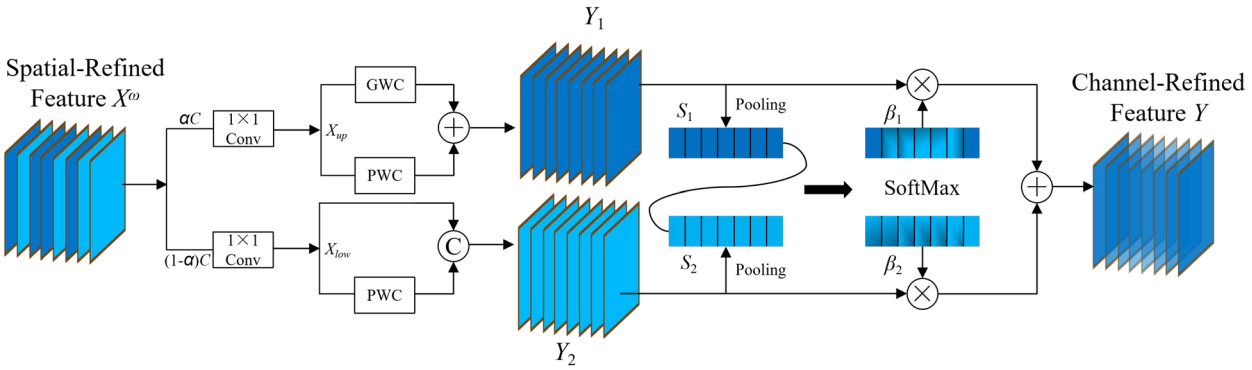


Fig. 3. The architecture of channel reconstruction unit.

$$\begin{cases} X_{11}^{\omega} \oplus X_{22}^{\omega} = X^{\omega 1}, \\ X_{21}^{\omega} \oplus X_{12}^{\omega} = X^{\omega 2}, \\ X^{\omega 1} \cup X^{\omega 2} = X^{\omega}. \end{cases} \tag{2}$$

While SRU optimizes spatial features, CNNs still suffer from channel redundancy, where multiple channels encode similar or redundant information. As shown in Fig. 3, the CRU is designed to compress redundant channels while preserving key feature representations. It follows a three-step process:

- Split Mechanism: Divides feature maps into two subsets, one for extracting high-level information and the other for preserving fine-grained details.
- Transform Mechanism: Applies lightweight convolutional operations to extract and refine features.
- Fuse Mechanism: Recombines processed features using an adaptive attention mechanism.

The input channels of X^{ω} are divided into two subsets: X_{up} is responsible for extracting high-level, rich features. X_{low} preserves low-level, shallow features. The split ratio α determines the percentage of channels allocated to each subset. Each subset undergoes different transformations. X_{up} is processed using group-wise convolution (GWC) and point-wise convolution (PWC) to extract high-level representations efficiently. X_{low} undergoes cheap 1×1 PWC to retain fine-grained details while minimizing computational cost. The outputs of the two subsets are as follows:

$$\begin{cases} Y_1 = M^{G} X_{up} + M^{P_1} X_{up}, \\ Y_2 = M^{P_2} X_{low} \cup X_{low}. \end{cases} \tag{3}$$

To ensure optimal feature utilization, CRU employs an adaptive fusion strategy. First, Global Average Pooling (GAP) is applied. Then, the upper and lower global channel-wise descriptors (S_1 and S_2) are stacked together. To determine the relative importance of each channel, a channel-wise soft attention operation is applied, generating the feature importance vectors β_1 and β_2 as follows:

$$\beta_1 = \frac{e^{S_1}}{e^{S_1} + e^{S_2}}, \beta_2 = \frac{e^{S_2}}{e^{S_1} + e^{S_2}}, \beta_1 + \beta_2 = 1 \tag{4}$$

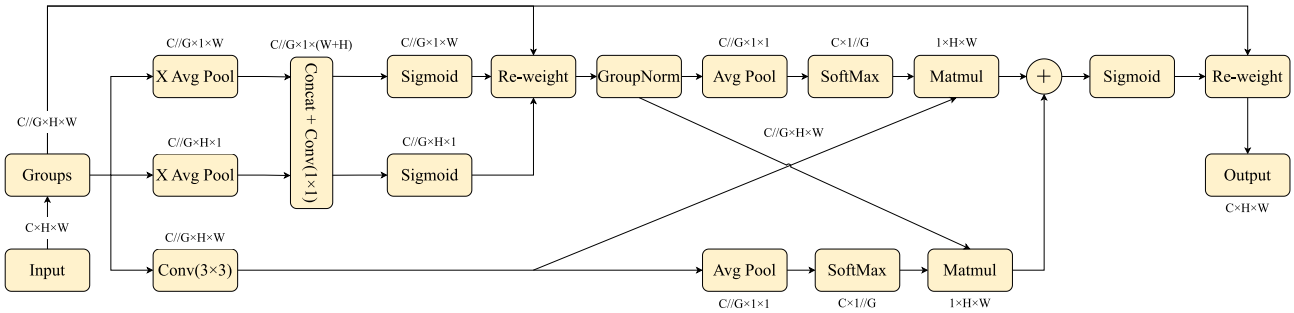
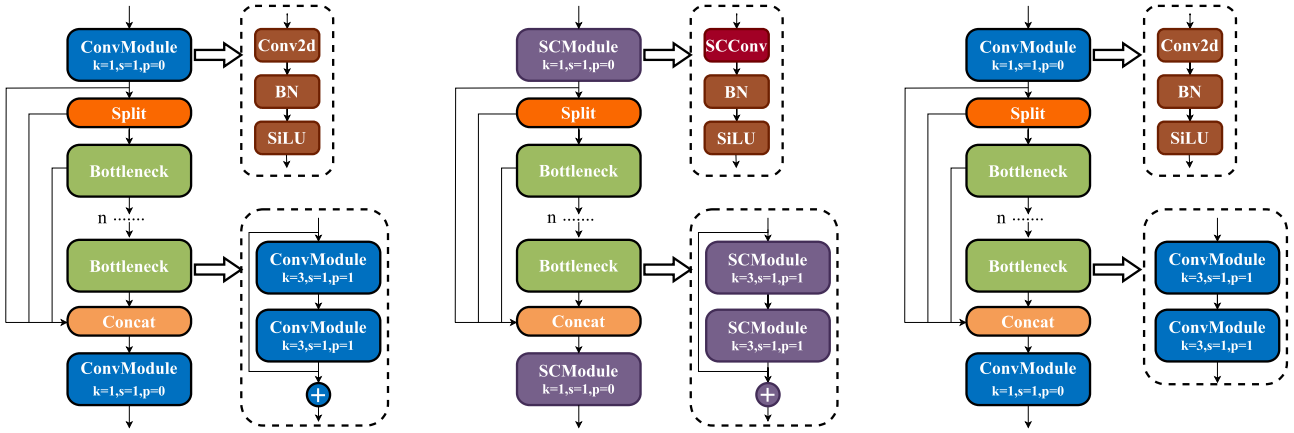


Fig. 4. The architecture of efficient multi-scale attention module.



(a) C2f in backbone (b) C2f SCConv (c) C2f in neck

Fig. 5. Comparison of C2f and C2f_SCConv.

Since $\beta_1 + \beta_2 = 1$, these values effectively represent a soft attention mechanism, dynamically adjusting the contribution of the upper and lower feature representations. The final output Y is obtained by a weighted sum of Y_1 and Y_2: $Y = \beta_1 Y_1 + \beta_2 Y_2$. This process ensures that the final output Y integrates information adaptively, emphasizing more important channel features while suppressing less significant ones.

EMA structure and mechanism

EMA enhances feature representation through channel grouping. This strategy divides features into independent subsets, balancing computational efficiency with information retention through parallel processing. As shown in Fig. 4, EMA employs three parallel branches: two 1×1 convolution paths for cross-channel interactions and one 3×3 path for spatial dependencies. The 1×1 branches concatenate features along the height dimension. They apply shared convolutions without dimensionality reduction and factorize the outputs. These outputs are processed by sigmoids and element-wise multiplication to enhance cross-channel interactions. Spatial attention maps are generated via global average pooling from both branches. Finally, the two spatial attention weights are aggregated and passed through a sigmoid function, capturing pixel-level pairwise relationships and enhancing global context awareness. The final EMA output retains the same size as input X, ensuring efficiency and seamless integration into modern architectures.

Design and integration of SCConv and EMA module

In the YOLOv8 backbone, the C2f module plays a crucial role in enhancing feature extraction. As shown in Fig. 5a, the backbone C2f splits inputs into 1–6 sub-branches for feature extraction: one keeps original features, others use bottleneck layers. However, prior studies have suggested that the standard C2f module in YOLOv8 can introduce redundant parameters/features and non-negligible computational overhead, which may slow down inference and yield inconsistent accuracy gains[26,27]. C2f has also been reported to be less effective in modeling long-range sparse features and in preserving fine details for small objects[28]. To address these challenges, as shown in Fig. 5b, the C2f_SCConv module replaces the standard Conv2d layers inside the ConvModule with SCConv while retaining the original C2f structural design. This modification enables a more effective decoupling of spatial and channel feature learning, strengthening spatial–channel interaction and improving local feature extraction. This is particularly beneficial for elongated cracks and small-scale defects under resource constraints. Additionally, SCConv optimizes the computational efficiency of the network by reducing the redundancy associated with conventional convolutions, leading to a more efficient backbone design.

As shown in Fig. 5c, the C2f module in the neck employs two sequential ConvModules. However, using C2f modules in both the backbone and neck can lead to redundant feature processing and suboptimal feature adaptation for the detection head. To optimize the feature fusion process in YOLOv8's neck, we replace the C2f module with the EMA module. EMA simplifies the feature fusion by employing three parallel routes to extract attention-weight descriptors from grouped feature maps, reducing redundancy while enhancing feature representation. At the same time, EMA integrates channel and spatial attention mechanisms to adaptively weight features across different levels. This improves coordination between high-level semantic information and low-level details, making it particularly effective for small object detection and complex scene understanding.

Experiments and results analysis

This section presents the experimental design used to identify an effective and accurate approach for road damage detection. We systematically compare several representative object detectors to select a strong baseline model. Based on this baseline, we integrate C2f_SCConv and EMA modules using different strategies. We then perform comprehensive quantitative, statistical, and visual analyses to evaluate the proposed variants and determine the optimal configuration. All experiments in this paper are conducted in the Google Colab cloud environment using an NVIDIA A100 GPU (40 GB) with CUDA 12.4, Python 3.11.11, and PyTorch 2.5.1.

Dataset

The benchmark dataset used in this study is RDD2022, released as part of the Crowdsensing-based Road Damage Detection Challenge (CRDDC'2022)[29]. It contains images collected from six countries: India, Japan, the Czech Republic, Norway, the United States, and China. The dataset includes 47,420 images, of which 38,385 are used for training, and provides 55,007 labeled instances across four damage categories: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20), and potholes (D40). It spans multiple road types and a wide range of weather and background conditions. Image resolutions range from 512 $\times$ 512 to 3650 $\times$ 2044 pixels, further increasing dataset diversity. While road damage patterns are broadly consistent across countries, differences in acquisition devices lead to noticeable variations in image quality and style. Figure 6 shows representative examples of each damage type. The images are captured using various devices, including smartphones, high-resolution cameras, and Google Street View, ensuring substantial variability in viewpoints and imaging conditions.

Evaluation metrics

We use *Precision*, *Recall*, *F1-score*, and *mAP* to evaluate the model's performance in identifying and localizing different types of road damage in images. The following formulas show that true positives (*TP*) occur when

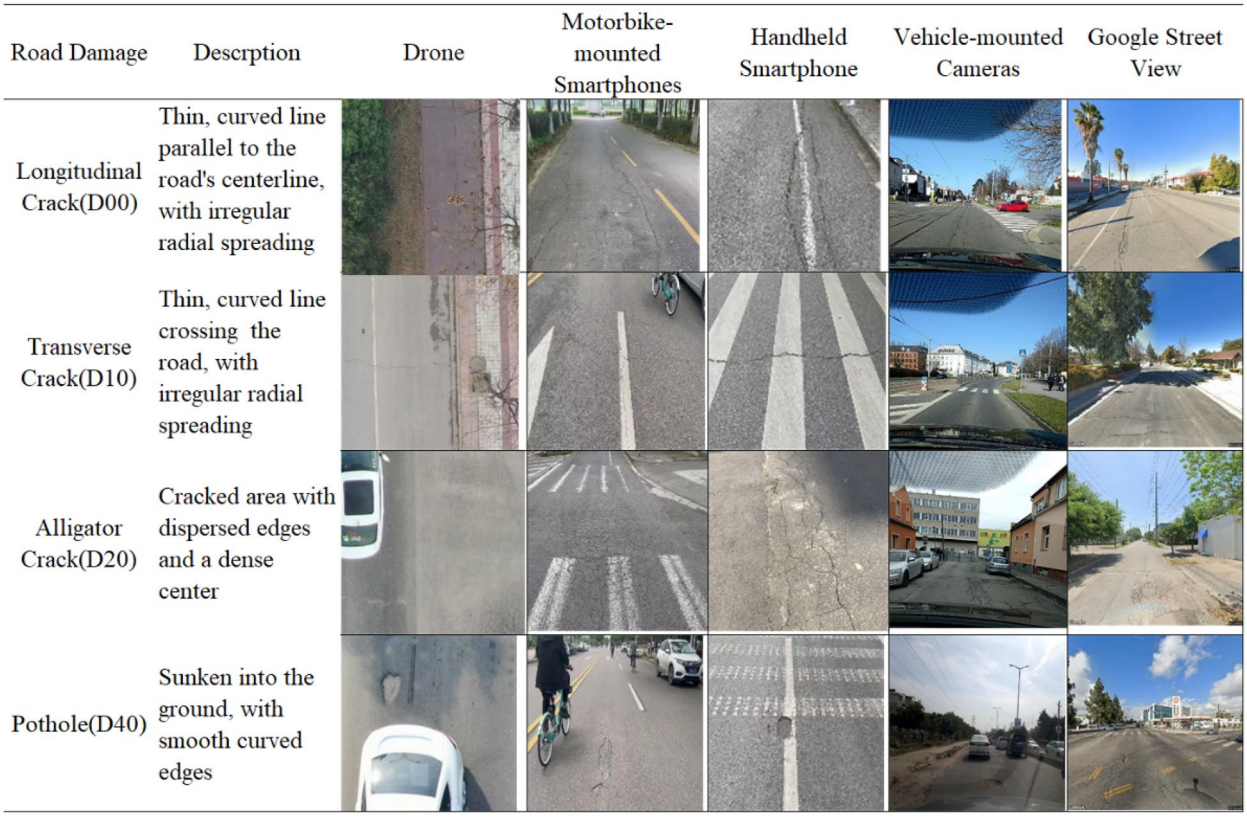


Fig. 6. Road damage examples captured with different devices.

the model correctly detects road damage. In contrast, false positives (*FP*) are incorrect detections in which the model identifies damage that does not exist. False negatives (*FN*) occur when the model fails to detect actual damage present in the image. The evaluation metrics are calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$\text{F1} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

mAP calculates the average precision (*AP*) of each class and then averages these values across all classes. *AP* is determined by plotting the precision-recall curve and calculating the area under this curve. *mAP@50 (%)* measures precision where a prediction is considered correct if its Intersection over Union (IoU) with the ground truth is greater than 0.5. *mAP@50–95 (%)*, on the other hand, calculates the average precision at multiple IoU thresholds (from 0.5 to 0.95, with steps of 0.05), providing a more comprehensive evaluation of model performance. Both metrics are commonly used to assess the accuracy and robustness of object detection models.

$$\text{mAP} = \frac{1}{\text{n}} \sum_{i=1}^{n} \text{AP}_i \tag{8}$$

Benchmark comparison

We conduct comparative experiments across a representative set of benchmark object detectors, including YOLOv5, YOLOv8, YOLOv11, RT-DETRv2, RT-DETRv3, and EfficientDet. These detectors provide multiple model scales and support the use of pre-trained weights to improve training efficiency and detection performance. Among the YOLO family, we include YOLOv5, YOLOv8, and YOLOv11 because they are widely adopted in both research and industry and are actively maintained by Ultralytics. The dataset is split into training, validation, and test sets with an 8:1:1 ratio, and Mosaic augmentation is applied to increase data diversity and robustness. Mosaic augmentation constructs a training sample by stitching four images into one, with random cropping and rescaling applied to each image region. All images are resized to 640 $\times$ 640 pixels for training. To ensure fair comparison, we keep the training protocol consistent across methods: 100 epochs, batch size 32, and an evaluation IoU threshold of 0.5. Training uses the SGD optimizer with momentum 0.937 and weight decay 0.0005, an initial learning rate of 0.01, and a linear learning-rate decay schedule over epochs.

Our evaluation across different model scales in Table 1 reveals remarkable trade-offs between model complexity and detection performance. At nano scales, models demonstrate reasonable performance with extremely low parameter counts and fast inference times. As we scale up to medium variants, we observe substantial improvements in detection accuracy. YOLOv8s and YOLOv11s share the high F1 score of 60.5% on this scale, though YOLOv11s achieves slightly better *mAP@50* (61.3% vs 60.9%). Among medium scale models, YOLOv5m delivers the highest precision of 69% within all the models, though it requires significantly more parameters (64M) than alternatives. YOLOv8m follows closely with a 68.9% precision and higher *mAP* while using only 25.8M parameters, offering a better efficiency trade-off. The large scale models further enhance detection performance. RT-DETRv3-R50 achieved the highest recall of 58.5% among all tested models. YOLOv8l achieved the highest *mAP@50* of 63.6% and an F1 score of 62.5%. Despite its 43.6M parameters and 164.8G FLOPs computational requirement, it maintains a competitive inference latency of 3.5 ms. Similarly, the extra large variants push the performance boundaries even further, with YOLOv11x achieving the highest *mAP@50–95* of 33.9%, albeit with substantially higher computational demands.

Analysis across different model scales indicates that YOLOv8l represents the strong balance between detection accuracy and computational efficiency. With leading metrics including recall (58.4%), *mAP@50* (63.6%), and an F1 score of 62.5%, combined with a reasonable inference latency of 3.5 ms, YOLOv8l outperforms other models in the critical balance of detection reliability and processing speed. While models like RT-DETRv3-R50 offer slightly lower latency (2.6 ms), YOLOv8l's superior detection performance justifies its selection as the baseline model for this study.

C2f_SCConv and EMA integration in YOLOv8l

Figure 7 presents a simplified network architecture of YOLOv8l with all network layers and their input-output dimensions marked. Analysis of the dataset reveals that road damage often appears under complex background with wide areas, so reliable detection benefits from deeper features that provide larger receptive fields and stronger contextual understanding. In our experiments, we integrated C2f_SCConv and EMA modules at different layers of the backbone and neck, respectively. We thoroughly considered the network architecture of PAN-FPN and implemented various combinations and replacements to comprehensively evaluate the integration efficiency across different levels of the network.

As shown in Table 2, using pretrained YOLOv8l weights during training effectively improves detection accuracy. Replacing C2f with C2f_SCConv in the backbone results in a significant reduction in both model size and computational complexity. Among the various replacement configurations, integrating C2f_SCConv at layer 6 produces the optimal combination of low latency (3.0ms in YOLO_SCC4) and considerable accuracy (63.1% *mAP@50*). This result aligns with our dataset observations, where defects typically present as extended cracks and compacted areas, benefiting from comprehensive feature extraction at deeper network levels.

Model	Precision (%)	Recall (%)	mAP@50 (%)	mAP@50–95 (%)	F1 (%)	Param. (M)	FLOPs (G)	Latency (ms)
Nano Models								
YOLOv5n	63.4	50.4	55.9	28.6	56.2	**2.5**	**7.1**	**0.9**
YOLOv8n	64.9	52.6	58.5	30.1	58.0	3.0	8.1	0.8
YOLOv11n	65.1	51.7	57.4	29.7	57.6	2.6	6.3	**0.9**
EfficientDet-D1	61.7	51.1	56.4	29.5	55.9	6.6	6.1	2.1
Small Models								
YOLOv5s	66.7	53.5	60.0	31.3	59.4	9.1	23.8	1.2
YOLOv8s	68.4	54.2	60.9	31.7	60.5	11.1	28.4	1.2
YOLOv11s	67.4	54.8	61.3	32.4	60.5	9.4	21.3	1.4
RT-DETRv2-R18	66.7	53.2	59.7	31.7	59.2	20.0	60.0	1.5
RT-DETRv3-R18	62.1	55.3	59.4	31.3	58.5	20.0	60.0	1.2
EfficientDet-D3	62.2	54.7	58.7	31.0	58.2	12.0	24.9	3.7
Medium Models								
YOLOv5m	**69.0**	55.8	62.6	32.8	61.7	64.0	25.0	2.1
YOLOv8m	68.9	55.6	62.7	33.2	61.5	25.8	78.7	2.3
YOLOv11m	68.8	55.1	62.5	33.2	61.2	20.0	67.7	2.5
RT-DETRv2-R34	66.6	54.7	61.0	32.2	60.1	31.0	92.0	2.0
RT-DETRv3-R34	65.8	55.2	61.1	32.4	60.0	31.0	92.0	1.9
EfficientDet-D4	62.5	56.0	59.9	31.6	59.1	20.7	55.2	7.2
Large Models								
YOLOv5l	66.0	57.8	62.5	33.1	61.6	134.7	53.1	3.2
YOLOv8l	67.4	58.4	**63.6**	33.7	**62.5**	43.6	164.8	3.5
YOLOv11l	67.6	56.6	62.7	33.7	61.6	25.3	86.6	3.2
RT-DETRv2-R50	66.5	55.2	61.2	32.4	60.3	42.0	136.0	2.9
RT-DETRv3-R50	66.6	**58.5**	63.2	33.4	62.2	42.0	136.0	2.6
EfficientDet-D5	65.5	56.0	61.1	32.0	60.4	33.7	130.0	10.3
Extra Large Models								
YOLOv5x	66.7	57.7	63.2	33.8	61.9	246.4	97.2	5.3
YOLOv8x	67.2	57.8	63.4	33.4	62.1	257.4	68.1	5.1
YOLOv11x	68.6	57.1	63.4	**33.9**	62.3	56.8	194.4	5.1
RT-DETRv2-R101	67.1	56.7	62.6	32.9	61.5	76.0	259.0	4.6
RT-DETRv3-R101	67.2	58.2	63.1	33.5	62.4	76.0	259.0	4.2
EfficientDet-D6	66.2	57.4	62.4	32.7	61.5	51.9	226.0	14.6

Table 1. Performance comparison of representative benchmark object detectors. Bold values indicate the best result for each metric among all compared models, corresponding to the highest value for Precision, Recall, mAP@50, mAP@50–95, and F1, and the lowest value for Params, FLOPs, and Latency.

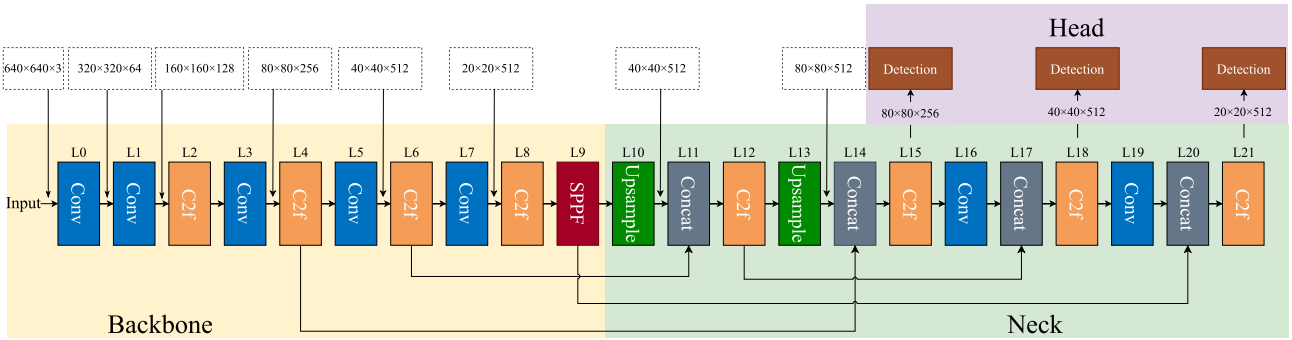


Fig. 7. A simplified architecture of YOLOv8l network.

As more C2f modules in the backbone are replaced, both model size and computational complexity decrease accordingly. However, inference time increases, and detection accuracy does not consistently improve, largely because the additional sequential split–transform–fusion operations introduced by C2f_SCConv can offset local savings but reduce end-to-end pipeline efficiency when applied at multiple stages. When C2f_SCConv is

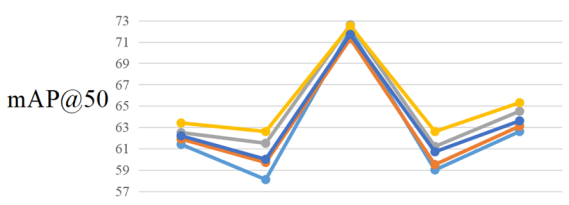
Model	Pretrained	C2f_SCConv	EMA	Param. (M)	FLOPs (G)	Latency (ms)	mAP@50 (%)	mAP@50–95 (%)
YOLO_SCC1	×	layer2,4,6,8	×	38.6	**141.3**	4.2	59.8	31.3
YOLO_SCC2	✓	layer2,4,6,8	×	38.6	**141.3**	4.2	60.6	32.0
YOLO_EMA1	×	×	layer12,15,18,21	43.6	165.3	4.8	62.7	32.6
YOLO_EMA2	✓	×	layer12,15,18,21	43.6	165.3	4.8	63.9	33.8
YOLO_EMA3	✓	×	layer15,18,21	43.5	165.1	4.7	64.5	34.1
YOLO_SCC3	✓	layer4	×	42.9	155.9	3.2	62.6	33.0
YOLO_SCC4	✓	layer6	×	40.8	155.8	**3.0**	63.1	33.4
YOLO_SCC5	✓	layer4,6	×	40.1	146.9	3.3	62.0	33.0
YOLO_SCC6	✓	layer4,15	×	42.6	151.4	3.5	62.0	32.7
YOLO_SCC7	✓	layer4,6,8	×	39.2	143.4	3.6	61.4	32.6
YOLO_SCC8	✓	layer6,12,18	×	**38.0**	146.8	3.3	62.9	33.3
YOLO_SE1	✓	layer4	layer15	42.9	156.1	3.3	62.8	33.1
YOLO_SE2	✓	layer4	layer15,18,21	42.9	156.4	3.5	64.6	34.1
YOLO_SE3	✓	layer6	layer12,15,18	40.8	156.1	3.2	64.1	33.8
YOLO_SE4	✓	layer6	layer15,18,21	40.8	156.3	3.2	64.8	34.2
YOLO_SE5	✓	layer6	layer12,15,18,21	41.1	156.6	3.5	64.2	33.8
YOLO_SE6†	✓	**layer4,6**	**layer15,18,21**	40.1	147.4	3.6	**65.3**	**34.5**
YOLO_SE7	✓	layer4,6	layer12,15,18,21	40.3	148.2	3.9	64.3	33.9
YOLO_SE8	✓	layer4,6,8	layer15,18,21	40.1	146.9	3.8	64.1	33.6
YOLO_SE9	✓	layer2,4,6,8	layer12,15,18,21	38.9	142.1	4.2	62.7	32.1
YOLOv8l_baseline	✓	×	×	43.6	164.8	3.5	63.6	33.7

Table 2. Comparison of different C2f_SCConv and EMA configurations. Bold values indicate the most favorable result for each metric among all compared configurations, corresponding to the highest value for mAP@50, mAP@50–95, and the lowest value for Params, FLOPs, and Latency. † indicates the final selected configuration.

extended to the neck region, detection accuracy decreases further despite reduced computational complexity. In contrast, selectively replacing only the three C2f modules closest to the detection head with EMA in the neck (as in YOLO_EMA3 with layers 15, 18, 21) yielded results exceeding the baseline accuracy (64.5% vs 63.6% *mAP@50*). EMA introduces a grouped multi-branch attention mechanism. Although this design helps enhance feature representation, it also incurs additional overhead from normalization, pooling, branch interaction, and attention reweighting, thereby resulting in increased model complexity and longer inference time. When C2f_SCConv and EMA are jointly integrated, the detector achieves higher accuracy. This gain is mainly attributed to their complementary roles: C2f_SCConv strengthens feature extraction, while EMA improves multi-scale fusion and feature re-weighting in the neck. Importantly, by applying these modules at PAN-FPN related stages, the network preserves efficient feature propagation and maintains fast inference with minimal additional overhead. YOLO_SE6 is selected for its superior overall accuracy–efficiency trade-off while maintaining near-baseline latency. YOLO_SE6 combines C2f_SCConv at backbone layers 4 and 6 with EMA at neck layers 15, 18, and 21, achieving a substantial performance improvement to 65.3% *mAP@50* and 34.5% *mAP@50–95* while maintaining computational efficiency.

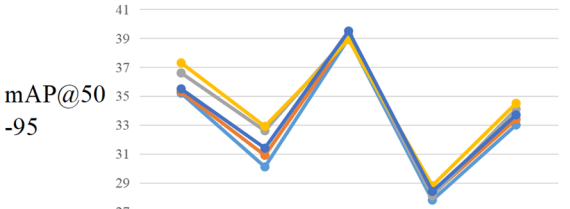
We analyze detection performance across all damage categories to better understand category-wise behavior, providing targeted guidance for model selection and refinement. As shown in Fig. 8, we evaluate some representative models across four critical detection metrics. The detection of elongated cracks (D00, D10) is highly susceptible to environmental influences, and the two types of cracks differ only in direction, making them easily confused. Among these, D00 (longitudinal crack) damage far outnumbers other categories in the dataset. D20 (alligator crack) consistently achieves the superior detection performance across models. This can be attributed to its distinctive texture patterns and typically larger damaged regions, which facilitate feature extraction and classification. By contrast, the inferior performance on D40 (pothole) is mainly related to limited samples, small target regions, and frequent occlusion. Although YOLO_SE6 does not achieve the highest precision in Fig. 8c, it attains the best overall mAP and recall performance. The joint use of C2f_SCConv and EMA improves sensitivity to weak and multi-scale damage cues by reducing feature redundancy and strengthening cross-spatial feature interaction. This is beneficial for overall performance, but it can also make the detector more responsive to ambiguous crack-like textures or complex background patterns, thereby increasing false positives in some categories and lowering precision at the default threshold. In contrast, YOLO_EMA3 introduces EMA only near the end of the neck, placing greater emphasis on reweighting the fused multi-scale features and suppressing noise. As a result, its predictions are relatively more conservative, which helps reduce false positives and thus leads to higher precision.

To further evaluate model generalization, we conduct validation experiments on UAV-PDD2023[30], a UAV-based pavement distress dataset with 2,440 images covering the four damage categories studied in this work.



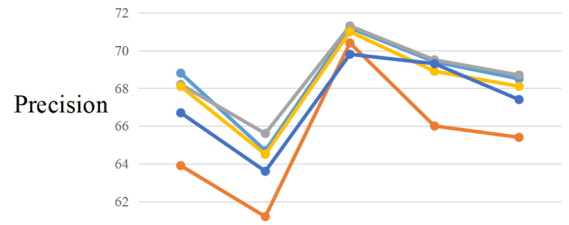
	D00	D10	D20	D40	Overall
YOLO_SCC3	61.4	58.1	72.1	59	62.6
YOLO_SCC4	61.9	59.7	71.3	59.5	63.1
YOLO_EMA3	62.5	61.5	72.6	61.2	64.5
YOLO_SE6	63.4	62.6	72.5	62.6	65.3
YOLOv8l	62.2	60	71.7	60.7	63.6

(a) mAP@50 performance on each road damage category



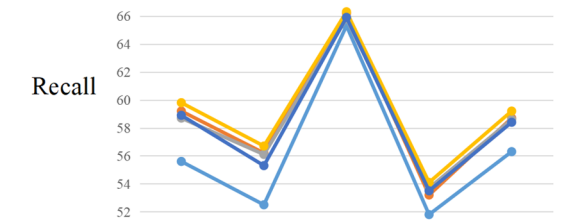
	D00	D10	D20	D40	Overall
YOLO_SCC3	35.2	30.1	39	27.8	33
YOLO_SCC4	35.3	30.9	39.2	28.2	33.4
YOLO_EMA3	36.6	32.6	39.2	28.1	34.1
YOLO_SE6	37.3	32.9	38.9	28.8	34.5
YOLOv8l	35.5	31.4	39.5	28.4	33.7

(b) mAP@50-95 performance on each road damage category



	D00	D10	D20	D40	Overall
YOLO_SCC3	68.8	64.7	71.2	69.4	68.5
YOLO_SCC4	63.9	61.2	70.4	66	65.4
YOLO_EMA3	68.2	65.6	71.3	69.5	68.7
YOLO_SE6	68.1	64.5	71	68.9	68.1
YOLOv8l	66.7	63.6	69.8	69.3	67.4

(c) Precision performance on each road damage category



	D00	D10	D20	D40	Overall
YOLO_SCC3	55.6	52.5	65.3	51.8	56.3
YOLO_SCC4	59.2	56.2	66.2	53.2	58.7
YOLO_EMA3	58.7	56.1	65.8	53.7	58.6
YOLO_SE6	59.8	56.7	66.3	54.1	59.2
YOLOv8l	58.9	55.3	65.9	53.5	58.4

(d) Recall performance on each road damage category

Fig. 8. Performance comparison across all damage categories.

Model	Precision(%)	Recall(%)	mAP@50(%)	mAP@50–95(%)	F1(%)	Latency(ms)
YOLOv8l	61.4	52.3	51.8	28.0	56.5	3.6
YOLOv11l	**63.2**	50.7	49.8	27.4	56.3	3.5
YOLO_SE6†	62.6	**54.2**	**56.5**	**29.2**	**58.1**	3.7
RT-DETRv3-R50	56.7	53.8	48.8	27.1	55.2	**3.1**

Table 3. Performance comparison on UAV-PDD2023. Bold values indicate the most favorable result for each metric among all compared models, corresponding to the highest value for Precision, Recall, mAP@50, mAP@50–95 and F1, and the lowest value for Latency. † indicates the final selected model.

This dataset is challenging for its aerial acquisition setting, leading to smaller target scales, larger scene coverage, and more complex background interference. Analysis of the dataset annotations shows that many damage instances are marked by small and highly overlapping bounding boxes. This annotation characteristic increases the difficulty of precise localization and makes missed detections and false detections more likely. Accordingly, the validation results in Table 3 indicate a clear performance decline for all compared models, while YOLO_SE6 shows the strongest overall detection capability, with the best Recall, *mAP@50*, *mAP@50–95* and F1. These results confirm its favorable generalization ability under more challenging conditions.

Visual analysis

In this study, we employ Grad-CAM++ to visualize and analyze the detection results. This visualization technique provides enhanced interpretability of the deep learning model's decision-making process by highlighting the regions of interest. As shown in Fig. 9, all evaluated models achieve high accuracy for single-category damage in scenes with minimal interference. For the D40 category, the highlighted regions overlap well with the damaged areas in the original images. YOLOv8l shows the strongest focus, with larger and clearer heatmap responses. However, the attention distribution becomes more critical when the background is cluttered or when multiple damage types appear together. In more challenging complex street scenario, the models show clear differences in attention patterns and localization behaviors. YOLOv11l is more affected by the surroundings, and its hotspots can drift away from the actual road damage. Although RT-DETRv3-50 shows concentrated heat responses over the damaged regions, it fails to correctly detect the corresponding damage instances. YOLO_SE6 maintains relatively high confidence and shows more consistent attention to the damaged areas. Some additional

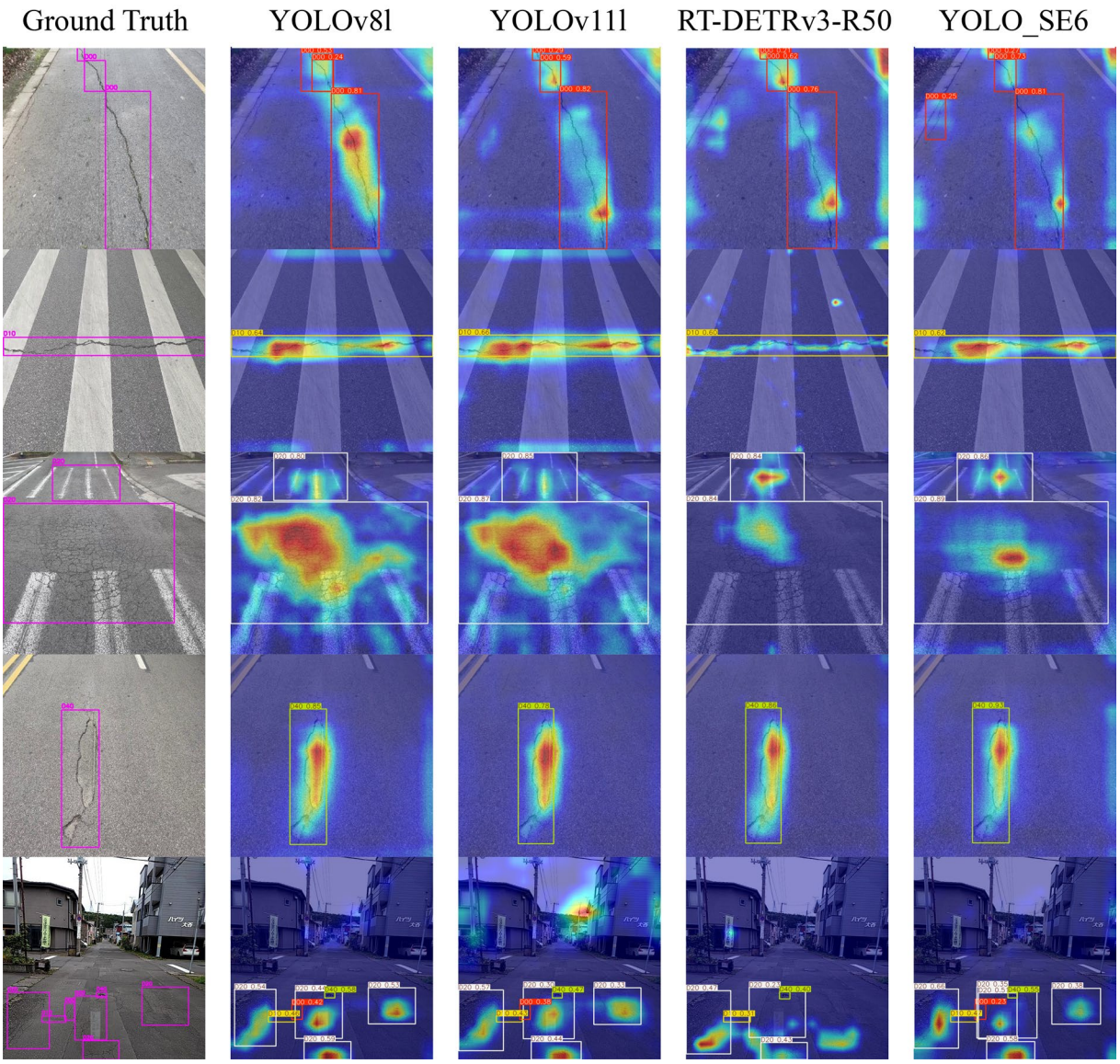


Fig. 9. Comparison of visual results using Grad-CAM++.

detections in the first and fifth row may correspond to unlabeled but visually evident damage areas, indicating higher sensitivity to weak and multi-scale damage cues, although they are counted as false positives under the annotation protocol. This observation further supports the precision analysis in Fig. 8c.

We further evaluate the robustness of YOLO_SE6 under challenging weather and illumination conditions. As shown in Fig. 10, the examples include wet pavement with water reflections, strong shadow interference, and low-light scenes. The proposed model can still localize road damage in these cases, although some redundant detections and inter-class confusion remain. In the first column, the model produces an additional D20 (alligator crack) detection covering a larger damaged area. In the second column, while the D20 (alligator crack) region is correctly detected, part of it is also predicted as D00 (longitudinal crack), indicating residual confusion between visually similar crack categories. In addition, the prediction confidence is more sensitive to illumination variations and tends to decrease in dim or backlit conditions. These results highlight the practical value of our design in maintaining reliable detection under diverse environmental conditions, while future improvements can focus on illumination robustness.

Conclusion and future work

This study presents a comprehensive approach for road damage detection by integrating SCConv and EMA modules into the YOLOv8 architecture. Based on extensive comparative experiments, we select YOLOv8l as the baseline for road damage detection, as it offers a strong balance between detection accuracy and computational efficiency. Our proposed YOLO_SE6 model, which incorporates C2f_SCConv at backbone layers 4 and 6 alongside EMA at neck layers 15, 18, and 21, improves detection accuracy with 65.3% *mAP@50* and 34.5% *mAP@50–95* while maintaining reasonable computational demands. Comprehensive visual analysis suggests



Fig. 10. Qualitative results under challenging weather and lighting conditions. The top row shows the ground-truth annotations, and the bottom row presents the corresponding detection results of YOLO_SE6.

that the enhanced model yields more discriminative feature representations and exhibits stronger generalization in complex scenes containing multiple damage types.

For future work, we plan to redesign the bottleneck structure in the C2f module to further optimize feature-extraction efficiency and improve detection performance under low-light conditions. We also plan to explore additional network architecture refinements, such as optimizing the neck-level feature fusion to better balance multi-scale representation and computational cost. We will further validate the proposed method on additional public datasets to better assess its generalizability. These efforts aim to address the remaining limitations of the current architecture and further improve detection performance in challenging road damage scenarios.

Data availability

The research data used for the experiments in this paper can be found as followings: RDD2022 dataset is available at: https://github.com/sekilab/RoadDamageDetector, UAV-PDD2023 dataset is available at: https://zenodo.org/records/8429208.

Received: 18 October 2025; Accepted: 15 April 2026
Published online: 20 April 2026

References

1. Manurung, E. H., Sawito, K., Satoto, A. & Tuanany, N. Analysis of the causes of road damage. *Civilla J. Tek. Sipil Univ. Islam Lamongan* **7**, 87 (2022).
2. Tsubota, T., Fernando, C., Yoshii, T. & Shirayanagi, H. Effect of road pavement types and ages on traffic accident risks. *Transp. Res. Procedia* **34**, 211–218 (2018).
3. Tang, Z., Chamchong, R. & Pawara, P. A comparison of road damage detection based on yolov8. In *2023 International Conference on Machine Learning and Cybernetics (ICMLC)*, 223–228 (IEEE, 2023).
4. Abdelwahed, S. H., Sharobim, B. K., Wasfey, B. & Said, L. A. Advancements in real-time road damage detection: A comprehensive survey of methodologies and datasets. *J. Real-Time Image Process.* **22**, 1–13 (2025).
5. Li, J., Wen, Y. & He, L. Scconv: Spatial and channel reconstruction convolution for feature redundancy. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 6153–6162 (2023).
6. Ouyang, D. et al. Efficient multi-scale attention module with cross-spatial learning. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5 (IEEE, 2023).
7. Redmon, J., Divvala, S., Girshick, R. & Farhadi, A. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, 779–788 (2016).
8. Liu, W. et al. Ssd: Single shot multibox detector. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*, 21–37 (Springer, 2016).
9. Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, 2980–2988 (2017).
10. Zhu, X. et al. Deformable detr: Deformable transformers for end-to-end object detection. arXiv preprint arXiv:2010.04159 (2020).
11. Tian, Z., Shen, C., Chen, H. & He, T. Fcos: Fully convolutional one-stage object detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, 9627–9636 (2019).
12. Girshick, R., Donahue, J., Darrell, T. & Malik, J. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, 580–587 (2014).
13. Li, X. & Zhang, Y. A lightweight method for road damage detection based on improved yolov8n. *Engineering Letters* **33** (2025).
14. Lin, Z. & Pan, W. Yolo-roc: A high-precision and ultra-lightweight model for real-time road damage detection. *Meas. Sci. Technol.* **37**, 035405 (2026).
15. Lan, X., Yang, Z. & Wang, H. Lightweight road distress detection model lrdd-yolov8 for efficient and accurate detection in complex scenarios. *Meas. Sci. Technol.* **36**, 075412 (2025).

16. Mao, J., Wang, M. & Liu, Z. Yolo-r: Road damage detector based on dynamic receptive fields and lightweight design: J. mao et al. *Signal, Image and Video Processing* **19**, 885 (2025).
17. Ye, L. et al. Yolov8n-sbp: A lightweight and efficient model for pavement distress detection. *J. Transp. Eng. Part B Pavements* **151**, 04025048 (2025).
18. Yu, A., Gao, Y., Xiong, Y., Liu, W. & She, J. Mobile-yolo: A lightweight yolo for road crack detection on mobile devices. *J. Adv. Comput. Intell. Intell. Inform.* **29**, 1443–1453 (2025).
19. Liu, Z., Jing, K., Yang, K., Zhang, Z. & Li, X. Efficient dense attention fusion network with channel correlation loss for road damage detection. *IET Intell. Transp. Syst.* **18**, 1747–1759 (2024).
20. Pan, W., Wang, X. & Huan, W. Real-time dynamic scale-aware fusion detection network: Take road damage detection as an example. *J. Real-Time Image Process.* **22**, 55 (2025).
21. Wang, N. et al. Deep learning network model for road pavement damage detection based on yolov8s-ges. *Transp. Res. Rec.: Transp. Res. Board* https://doi.org/10.1177/03611981241312224 (2025).
22. Liu, Z. et al. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, 10012–10022 (2021).
23. Irsal, R. B. P., Utaminingrum, F. & Ogata, K. Swin transformer adaptation into yolov7 for road damage detection. *Bull. Electr. Eng. Inform.* **13**, 2527–2536 (2024).
24. Roy, A. M. & Bhaduri, J. Densesph-yolov5: An automated damage detection model based on densenet and swin-transformer prediction head-enabled yolov5 with attention mechanism. *Adv. Eng. Inform.* **56**, 102007 (2023).
25. Zhao, G., Zhang, W., Sun, R. & Wei, T. Road crack detection based on improved rt-detr. *Signal Image Video Process.* **19**, 583 (2025).
26. Chen, X., Jiao, Z. & Liu, Y. Improved yolov8n based helmet wearing inspection method. *Sci. Rep.* **15**, 1945 (2025).
27. Hua, G. et al. An enhanced yolov8-based bolt detection algorithm for transmission line. *IET Gener. Transm. Distrib.* **18**, 4065–4077 (2024).
28. Meng, L. et al. Rla-yolo: A lightweight and accurate method for real-time steel plate defect detection. *Digit. Signal Process.* **168**, 105700 (2026).
29. Arya, D., Maeda, H., Ghosh, S. K., Toshniwal, D. & Sekimoto, Y. Rdd 2022: A multi-national image dataset for automatic road damage detection. *Geosci. Data J.* **11**, 846–862 (2024).
30. Yan, H. & Zhang, J. Uav-pdd2023: A benchmark dataset for pavement distress detection based on uav images. *Data Brief* **51**, 109692 (2023).

Acknowledgements

We are grateful for all the cooperation and support from the department of computer science of MSU.

Author contributions

Z.T. and R.C. conceived and designed the study; Z.T. performed the experiments and analyzed the data; Z.T. drafted the manuscript; H.W. validated the results and revised the manuscript; R.C. provided funding support. All authors reviewed and approved the final manuscript.

Funding

This research project was financially supported by Mahasarakham University.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to R.C.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026